Frame Annotation of Legal Reporting Obligations: A Dataset and Baseline

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Abstract

Reporting obligations are among the most pervasive instruments of European Union regulation, yet locating and interpreting them in legal text remains largely manual. We present a frame-semantic resource for this task: a domain-specific inventory of FrameNet frames tailored to reporting obligations and centred on the reporting act, a gold-standard dataset that annotates the corpus of reporting requirements of Palmirani et al. provision by provision, and a baseline annotation system that reproduces the same annotations automatically using a large language model. The annotations are serialised as interoperable linked data using the NIF standard, with frames linked to Framester URIs and provisions to the European Legislation Identifier, so that the resulting frame graphs can feed legal knowledge graphs and compliance analysis.

1 Introduction

Reporting obligations are among the most pervasive instruments of European Union regulation. Across domains as diverse as banking, insurance, financial markets, data protection, environmental protection and product safety, EU legislation routinely requires regulated actors to transmit specified information to a competent authority, frequently on a recurring basis and within strict time limits. A credit institution must file prudential reports to its supervisor at regular intervals; a data controller must notify the supervisory authority of a personal-data breach without undue delay and, where feasible, within seventy-two hours; an operator must report emissions or incidents to a designated agency. Despite their diversity, such provisions share a recognisable semantic structure: an obligated party, an act of submitting or notifying, a recipient authority, the information to be reported, and temporal parameters expressing deadlines, periodicity or reference points.

The volume and heterogeneity of these obligations has itself become a regulatory concern. Scattered across hundreds of legal acts and drafted in dense natural language, reporting requirements are costly to identify, monitor and keep current, which has motivated efforts both to rationalise them at the policy level and to represent them formally. On the formal side, Palmirani et al. [7] propose an ontology dedicated to reporting requirements in European legislation, modelling the actors involved, the content to be reported and the temporal conditions that govern it. Populating such a model, however, presupposes the ability to locate and interpret the relevant provisions in the legal text itself, a step that remains largely manual.

Frame semantics offers a natural intermediate representation for this step. Prior work on contract law, for example ContractFrames [6], has shown that frame-based structures can bridge natural-language legal text and formal logic by capturing events and their participants in a normalised form. We pursue an analogous strategy for regulatory reporting: we annotate the provisions that impose reporting duties with FrameNet [1] frames, making explicit who must report what, to whom, and when, and we serialise the result as interoperable linked data so that it can feed ontologies such as the one above.

In this paper we make the following contributions. First, we define a domain-specific inventory of FrameNet frames tailored to reporting obligations, adapting general-purpose frames so that their frame elements align with the participants and temporal parameters characteristic of this domain. Second, we present a gold-standard dataset of EU legal provisions annotated with this inventory, together with the annotation methodology. Third, we provide a baseline annotation system built on a large language model and evaluate it against the gold standard. Finally, we discuss applications of the resource to legal knowledge graphs and compliance analysis.

2 State of the Art

FrameNet. FrameNet [1] is a lexical database of English grounded in frame semantics [2], the theory that words evoke structured conceptual frames representing stereotypical situations, events or relations. Each frame defines a set of participants called Frame Elements (FEs), which may be core (semantically obligatory) or non-core (peripheral). FrameNet 1.7 contains over 1,200 frames, 13,000 lexical units and 200,000 annotated example sentences. Frames are interconnected through ontological relations such as *Inheritance*, *Using*, *Subframe* and *Perspective_on*, forming a rich semantic network that supports both lexicographic and computational applications.

Framester. Framester [3] is a large-scale RDF/OWL linked data hub developed at CNR and Università di Bologna that integrates FrameNet with WordNet, VerbNet, BabelNet, DBpedia, Yago and the DOLCE-Zero foundational ontology into a single strongly connected knowledge graph. Frames are formalised as OWL classes, frame elements as properties, and lexical units as linked data resources, enabling full-fledged SPARQL querying and OWL reasoning over frame-based knowledge. Framester exposes a public SPARQL endpoint[1] and its data is available on GitHub.[2] In this work, Framester URIs are used to serialise frame annotations as interoperable Turtle RDF using the NIF standard [4].

2.1 Frame Semantic Parsing

Frame semantic parsing is the task of automatically producing FrameNet-style analyses of text, conventionally decomposed into three interdependent subtasks: frame identification (selecting the frame evoked by a target), argument identification (delimiting the spans that fill its roles) and role classification (labelling those spans with frame elements). A representative recent approach is the double-graph framework of Zheng et al. [8], whose KID model (Knowledge-guided Incremental semantic parser with Double-graph) recasts parsing as incremental graph construction. KID couples two graphs: a *Frame Knowledge Graph*, a static heterogeneous graph that encodes frames and their frame elements together with the ontological relations declared in FrameNet, and a *Frame Semantic Graph*, the per-sentence structure built up incrementally from the text. By letting knowledge-enhanced frame and frame-element representations from the former guide the construction of the latter, the model strengthens the interactions between the three subtasks and the dependencies among arguments that pipeline systems treat in isolation, reporting improvements of up to 1.7 F_1 over the previous state of the art on two FrameNet benchmarks. This separation between an ontological knowledge layer and an instance-level semantic graph is conceptually close to the representation we adopt, although our baseline delegates the parsing itself to a large language model rather than learning a dedicated graph-construction model (Section 4).

2.2 Frame Semantics in Legal Text

The application of frame semantics to legal language has a smaller but growing body of work. Closest to our setting, ContractFrames [6] applies a frame-based representation to contract law, extracting events related to the status of a purchase contract into frames and mapping them to logic clauses of the PROLEG legal-reasoning system through a dedicated Contract Workflow Ontology; it thereby uses frames as an intermediate layer between natural-language contractual text and formal logic, a role analogous to the one we envisage for reporting obligations. More broadly, the enrichment of legal documents with interoperable linguistic annotations has been demonstrated at platform scale by the Lynx project [5], whose service platform processes, enriches and analyses documents from the legal domain and represents the resulting linguistic and semantic annotations as linked data using the NIF standard. Our work follows the same linked-data annotation philosophy, narrowing the focus to FrameNet frames for reporting obligations and serialising them through Framester URIs.

Most directly related to our domain, Palmirani et al. [7] propose a Reporting Requirements Ontology for European legislation, a conceptual model that formalises the obligations imposed on regulated actors to submit information to authorities, capturing the parties involved, the content to be reported, and the temporal conditions such as deadlines and periodicity. Whereas that work operates at the ontological level, defining the classes and properties needed to represent reporting duties, our contribution is complementary and operates at the linguistic level: we annotate the textual provisions themselves with FrameNet frames, yielding the surface evidence from which such ontological structures can be populated. The frame inventory we adopt is in fact designed so that its frames and frame elements align with the participants and temporal parameters of reporting obligations, providing a natural bridge towards ontologies of this kind.

3 Dataset

The dataset is built on top of the corpus assembled by Palmirani et al. [7] for their Reporting Requirements Ontology. It consists of 93 reporting requirements drawn from European Union legislation across a range of regulatory domains — among them prudential banking supervision, securities markets, central counterparties and product approval — each one a self-contained provision (typically an article paragraph or subparagraph) that imposes a duty to transmit specified information to a competent authority. Reusing this corpus means that our frame annotations are grounded in provisions whose actors, reported content and temporal conditions have already been studied and modelled at the ontological level, so that the linguistic layer we add aligns directly with an existing conceptual representation of the same texts.

Each provision is manually annotated with the tailored subset of FrameNet [1] frames defined for reporting obligations — with TELLING as the central reporting act, alongside REQUEST, RECEIVING and the conditional and temporal frames — yielding, for every reporting requirement, a small *frame graph*: the set of frames it evokes together with the participants and temporal parameters that fill their

[1]https://w3id.org/framester/sparql
[2]https://github.com/framester/Framester

frame elements. A provision such as *"Within* 270 *days of receipt of the conclusions of the evaluation, the Agency shall prepare and submit to the Commission an opinion"* thus surfaces a TELLING frame (the obligated party, the recipient authority and the reported content) anchored to the temporal frames that express its deadline. The annotation was carried out by a single annotator, Célestine Lecat, a linguist rather than a legal expert; frames and frame elements are therefore assigned on linguistic, frame-semantic grounds — according to the predicates and arguments actually realised in the text — rather than through legal interpretation of the underlying obligation. This keeps the annotation close to the surface evidence and reproducible from the wording alone, while leaving the legal characterisation of each duty to the complementary ontological layer.

The resulting annotations are serialised differently from the ontology population of Palmirani et al. Rather than instantiating domain classes, we publish them as interoperable linked data following the NIF standard [4]: every provision is a `nif:Context` carrying its verbatim text and character offsets, and each target and frame element is a NIF string span with explicit begin and end indices into that context. Frames are typed with their Framester [3] URIs, linking every annotation into the wider FrameNet/WordNet/VerbNet knowledge graph, and each provision is related through the European Legislation Identifier (ELI) to the act it is drawn from. The corpus is distributed as Turtle, so that the frame graphs can be queried with SPARQL and joined, via their Framester and ELI links, both to the Framester ecosystem and to the Reporting Requirements Ontology.

4 Baseline Annotation System

To accompany the dataset we provide a baseline annotation system that produces the same kind of frame graphs automatically. In keeping with its role as a baseline, it deliberately avoids any task-specific training: rather than learning a dedicated graph-construction model in the manner of Zheng et al. [8], it delegates frame identification and role labelling to a general-purpose large language model steered entirely by an instruction prompt, and reserves all character-level bookkeeping for a deterministic post-processor. It thus establishes a simple, reproducible reference point against which both the gold standard and future dedicated parsers can be compared.

4.1 System Description

The system is a small Java web application exposing a single annotation endpoint. A provision submitted through the endpoint passes through a three-stage pipeline: prompting a large language model, resolving the returned text spans to character offsets, and serialising the result as linked data.

The first stage is built on Claude (Anthropic), specifically the Claude Sonnet model (identifier `claude-sonnet-4-6`), queried through the Messages API in a zero-shot, instruction-only setting — the prompt contains no annotated examples. A fixed system prompt encodes the tailored frame inventory: for each frame it gives a one-line definition together with its core and non-core frame elements, and it states the annotation rules that the gold standard follows. The salient rules are that the target must be the content word that lexically evokes the frame (modal auxiliaries such as *shall* or *must* are never targets, only the main verb they modify), that every frame element must be copied as an exact substring of the provision rather than paraphrased, and that elements incorporated into the target itself are flagged rather than given a span. The inventory is applied in two passes: the model first exhaustively annotates *every* TELLING and REQUEST frame in the provision — so that a passage with separate *inform* and *disclose* clauses yields two distinct frames — and then adds the most relevant remaining temporal and conditional frames. The model returns a JSON structure listing, for each frame, its name, its target and its frame elements as verbatim text spans.

Asking the model only for the text of each span, and not for numeric offsets, isolates the annotation from the well-known unreliability of language models at character counting. The second stage therefore resolves these spans deterministically against the original provision. Each target is located in the text and used to delimit the sentence that contains it; the frame elements of that frame are then searched only within that sentence, and, where a span occurs more than once, the occurrence closest to the target is chosen, with a preference for whole-word matches. This positional resolution prevents a short element such as the pronoun *it* from binding to an unrelated earlier occurrence, and a per-target cursor ensures that a verb repeated across several reporting acts maps each frame to a successive occurrence. The resolved frames are finally ordered by their relevance to reporting duties, with TELLING first.

The third stage serialises the annotated provision as described in Section 3: frames are emitted with their Framester URIs and the target and frame-element spans as NIF strings carrying their begin and end indices, producing Turtle that is interoperable with the rest of the dataset. The same annotation is also available as raw JSON, and a lightweight web interface (Figure 1) renders the frames over the provision text and offers both serialisations for download, so that automatic annotations can be inspected and exported in the same format as the human-annotated gold standard.

4.2 Evaluation

5 Applications

Because each provision is annotated and published as interoperable linked data, the resource is intended less as an end in itself than as a bridge between the natural-language text of reporting obligations and the formal representations that legal and compliance systems operate on. We outline two application areas that motivate the design.

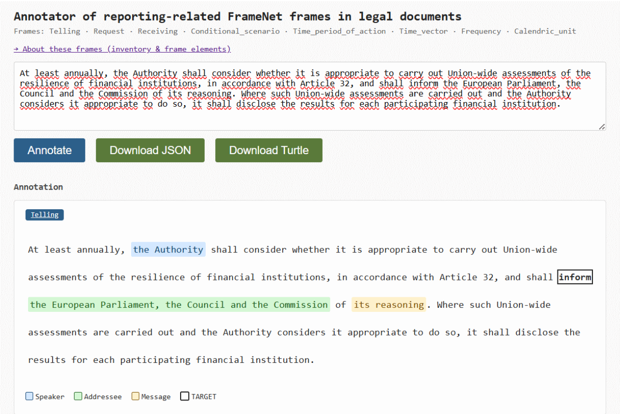

Figure 1: The web demo of the baseline annotation system. A provision is annotated and the detected frames are rendered over its text, with the target and frame-element spans highlighted; the JSON and Turtle serialisations can be downloaded.

5.1 Legal Knowledge Graphs

The frame graphs provide the surface evidence from which legal knowledge graphs can be populated. Each annotated provision already makes explicit, in a normalised form, the participants and temporal parameters of a reporting duty — who must report, to whom, what is reported, and under which deadlines, periodicity and conditions — which are precisely the elements that the Reporting Requirements Ontology of Palmirani et al. [7] models as classes and properties. Sharing the same underlying corpus, the frame annotations can therefore be mapped onto that ontology to help populate it, the linguistic layer supplying the textual anchors for the conceptual one. Beyond a single ontology, typing frames with Framester [3] URIs links every annotation into a wide linguistic knowledge graph that integrates FrameNet with WordNet, VerbNet and DBpedia, while the European Legislation Identifier connects each provision to the act it belongs to. The frame graphs are thus queryable with SPARQL and joinable across resources: one can ask, for instance, for all provisions that evoke a TELLING frame whose recipient is a particular authority, or that attach a recurring FREQUENCY to a reporting act.

5.2 Compliance Analysis

The same explicit structure supports compliance analysis. Identifying, monitoring and keeping reporting requirements current is costly precisely because they are scattered across many acts and expressed in dense prose; making their who–what–whom–when structure machine-readable is a step towards automating that work. The temporal frames isolate the conditions a compliance system must track — TIME_PERIOD_OF_ACTION and TIME_VECTOR for deadlines, FREQUENCY and CALENDRIC_UNIT for periodicity — while CONDITIONAL_SCENARIO captures the circumstances under which a duty is triggered, and the TELLING and RECEIVING frames identify the obligated party and the recipient authority. A repository of provisions annotated in this way could underpin tasks such as deadline tracking, obligation inventories for a given regulated actor, and the detection of changes to reporting duties as legislation evolves.

6 Conclusion

We have presented a frame-semantic resource for legal reporting obligations: a domain-specific inventory of FrameNet frames centred on the reporting act, a gold-standard dataset built on the corpus of Palmirani et al. [7] and annotated provision by provision into interoperable linked data, and a baseline annotation system that reproduces the same frame graphs automatically using a large language model. By serialising the annotations with NIF [4] and Framester [3] URIs and linking them through the European Legislation Identifier, the resource is designed to feed legal knowledge graphs and compliance analysis rather than to stand alone.

Several directions remain open. The most immediate is to complete the manual annotation of the corpus and to evaluate the baseline against it, both on frame identification and on frame-element resolution. Beyond that, the per-provision frame graphs invite an explicit modelling of the relations between co-occurring frames — a deadline attached to a reporting act, a condition governing it — in the spirit of the double-graph parsing of Zheng et al. [8], and a tighter alignment with the Reporting Requirements Ontology and with event vocabularies in the wider linked open data cloud.

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